DATA7201 Data Analytics at Scale - Project Report

Examining Non-Government Aligned Election Spending

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Abstract

This report examines non-government advertisers on the Facebook ad platform around the May 2022 federal election in order to determine the existence of American-style political action committee (PAC) advertisers in Australian elections.

Facebook ad data surrounding the election was ingested and preprocessed using PySpark on YARN. A pandas UDF tagged candidate, party, and government advertising via token matching against the Australian Electoral Commission candidate roll. The remaining ads were clustered with Latent Dirichlet Allocation and hand-labelled by category to exclude non-political content.

The corpus was then analysed in two ways: a deterministic classifier based upon ad spending before and after the election, and a machine learning classifier using hashtags to create a training dataset for a logistic-regression text classifier. Neither metric alone was sufficient to properly classify advertisers.

PAC-style advertisers were defined as those with strong pre-election spend relative to post-election spend combined with election-focused content in the weeks before polling day. Combining the two classifiers as a quadrant filter on these criteria identified a defensible list of Australian PAC equivalents.

A number of advertisers were subsequently found who matched the PAC-style advertisers seen in the United States, including Smart Voting, the Australian Education Union, Thrive By Five, the Australian Christian Lobby, GetUp!, ABC Friends, and It's Not A Race.

Contents

Word Count

The following is a summary of the word count this report, as generated by `texcount`.

Section	**Words**
Introduction	206
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Total	**1,601**

Table 1: Word count breakdown.

1 Introduction

Political Action Committees (PACs) are an American campaigning vehicle that raise money for political advertising outside of the existing political structure. Australia does not formally have PACs. Yet independent, third-party advertisers spend millions of dollars on advertising during Australian elections. Identifying these informal Australian "PACs" is important for political-advertising transparency, and for finding the boundary between advertisers who are campaigning on specific issues and advertisers who are campaigning on behalf of political parties with private funding.

The Facebook ad platform is attractive for advertisers as it provides access to a large viewer base with minimal effort. Traditional methods such as door-knocking and physical advertising are not required. The Facebook Ad Library is also a convenient data source for finding Australian PAC equivalents. A dataset was provided covering four years of political advertising in Australia, from 2020 to 2024. Although this corpus covered a range of political events, this report focuses on the 2022 Australian federal election.

At approximately 320 GB of JSON files, the dataset cannot easily be analysed on a conventional workstation. A distributed analytics framework is required. PySpark (Zaharia et al., 2016) was used for its lazy DataFrame API (Armbrust et al., 2015), and Spark MLlib (Meng et al., 2016) was used for both the unsupervised topic modelling (LDA) and the supervised text classifier (logistic regression).

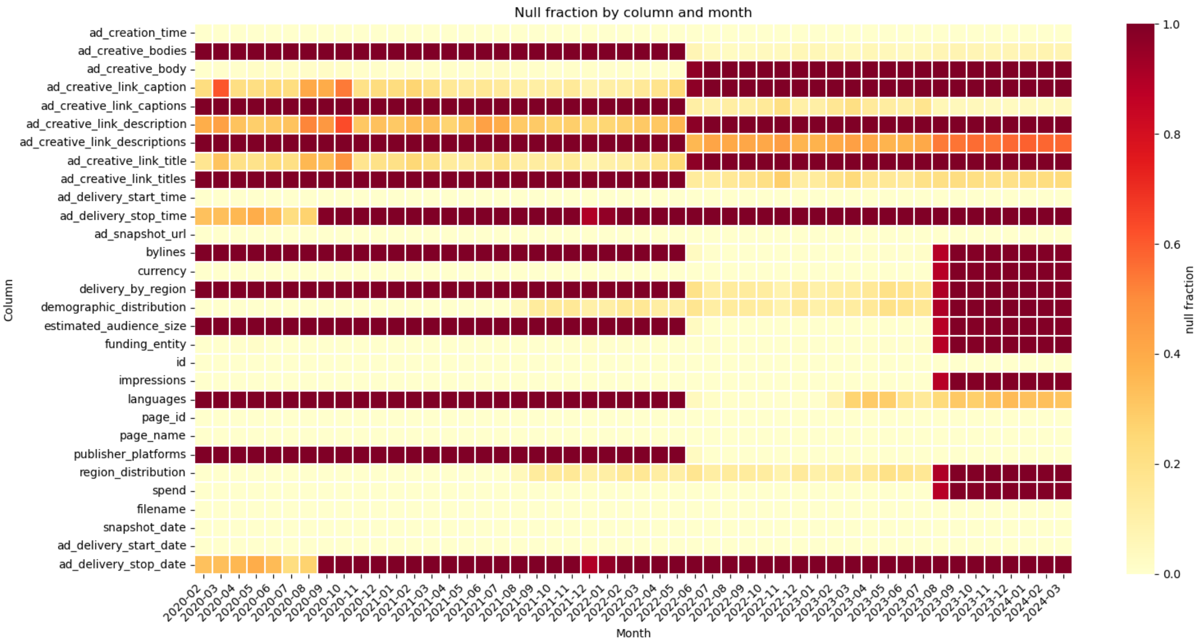
2 Dataset Analytics

2.1 Data ingestion and preparation

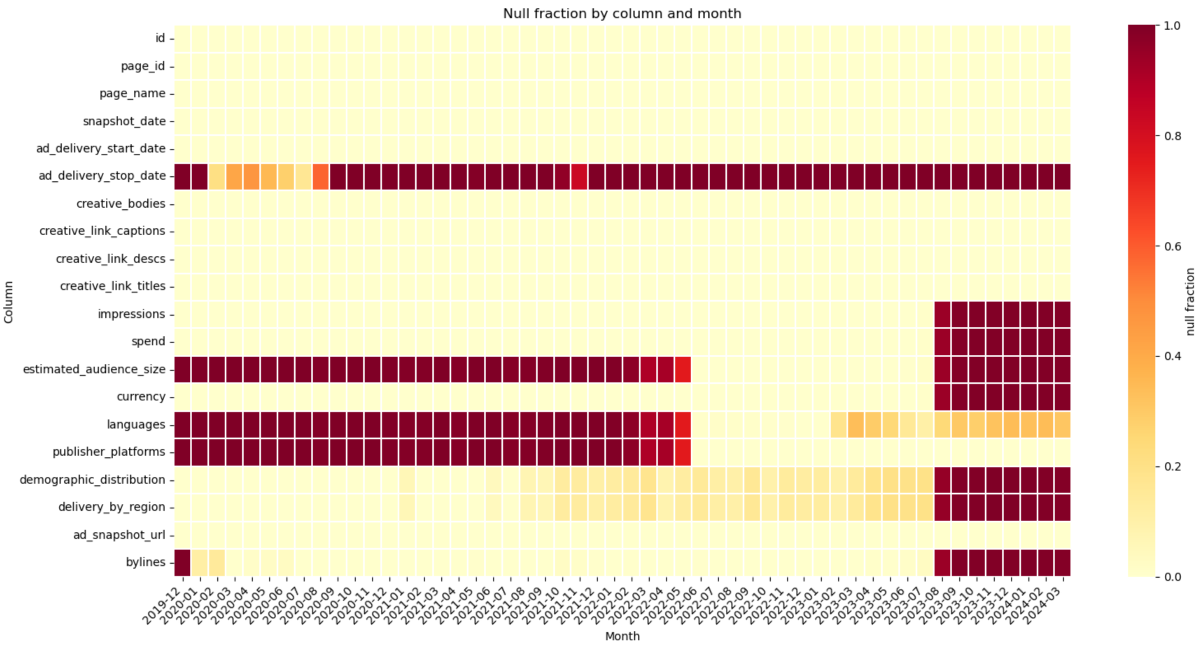
The data consisted of approximately 4,500 JSON files. These files contained around 40 million rows of data covering the period from February 2020 to March 2024. Each file was a snapshot of all Australian political advertisements that were active at the time the snapshot was taken. The same ad could therefore appear in many files. Each row represented a single advertisement with around 25 columns covering the ad purchaser, the Facebook page, the creative text, the spend, the impressions and various other fields. Some schema changes occurred in June 2022 and August 2023, with several key fields becoming sparse thereafter.

A null-fraction analysis was performed in order to visualise the schema drift. Automatic schema inference was used to collect the superset of all fields across the dataset. The per-column null fraction was then computed on a monthly basis and plotted as a heatmap (Figure 1). Columns were then merged to give a consistent schema across the window. Some fields were stored as nested structs of string-typed lower and upper bounds. These were cast to numeric values and replaced with their midpoint.

Deduplication was performed by sorting the snapshots chronologically per ad ID. Each row was then tagged with an `ad_seq_no` using a `row_number()` window operation, where a value of 1 indicates the most recent snapshot. The snapshot filenames used inconsistent date conventions and these required special handling. The cleaned data was then windowed to six months either side of the May 2022 federal election and written to HDFS as Parquet. This reduced the corpus from approximately 40 million rows down to 5.7 million rows.



(a) Before schema correction.



(b) After schema correction.

Figure 1: Null-fraction heatmaps (column × month) before and after the schema coalesce step. The raw view (top) shows the June 2022 singular-to-array transition and the August 2023 breakage clearly. The cleaned view (bottom) has the duplicate fields merged and a single consistent schema across the window. Hot colors show a high percentage of null values for that month/column.

2.2 Identifying government advertising

To focus on third-party organisations the government advertising had to be removed. This included both political-party advertisements and candidate advertisements. The 2022 House of Representatives and Senate candidate rolls were sourced from the Australian Electoral Commission. A straight join was not effective because page names can be misrepresentative and the byline can be paid by an aide. Thus a pandas UDF was used to apply three rules in order with the first match winning, summarised in Table 2.

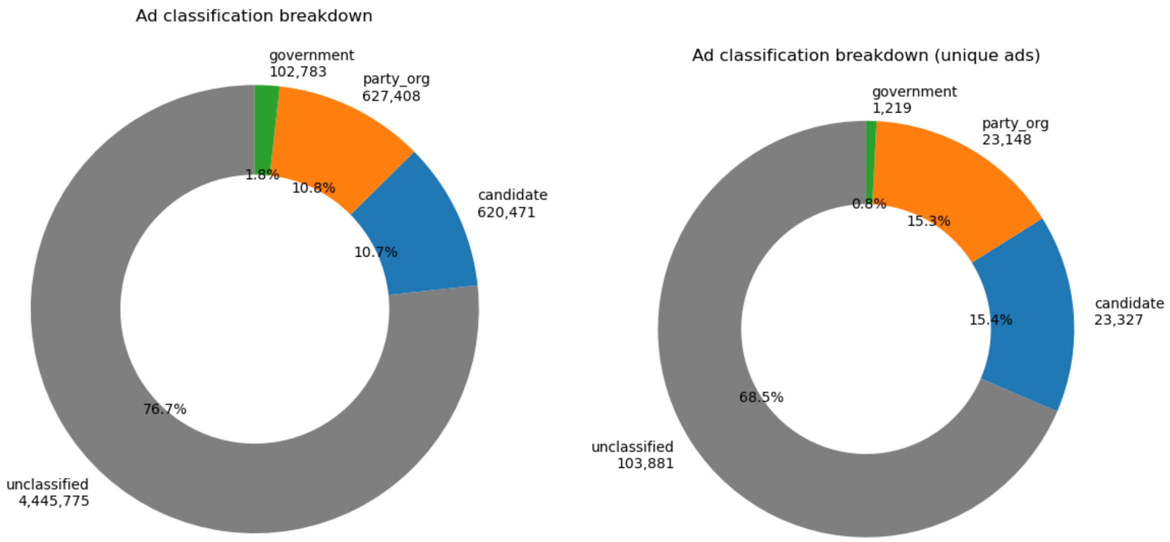
Rule	**Logic and output**
Candidate name	The candidate's tokenised name (surname plus first given name) is a subset of either the page-name tokens or the byline tokens.
Party byline signature	A tokenised party signature from `aec_parties.csv` is a subset of the byline tokens, e.g. {liberal, national} → LNP. Signatures are tested longest-first so LNP matches before LP.
Government byline signature	The byline tokens contain one of {department}, {australian, government}, or {australian, electoral, commission}.

Table 2: Three rules applied by the byline-matching pandas UDF in order, first match wins.

Every ad was thus tagged according to its nature; ads tagged `null` (no match against any of the three rules) formed the third-party corpus of interest. The row counts are shown in Table 3, with the match-type composition visualised in Figure 2 and the weekly spending pattern by match-type in Figure 3.

match_type	**total rows**	**unique ads**
null	4,511,742	105,814
party_org	649,857	23,932
candidate	536,232	20,644
government	98,660	1,188

Table 3: Rows and unique ads after government, party, and candidate classification.



(a) All rows (per snapshot). (b) Unique ads (`ad_seq_no`=1).

Figure 2: Match-type distribution by total rows (left) and by unique ads after deduplication (right). The `null` bucket dominates by both measures and forms the third-party corpus carried forward.

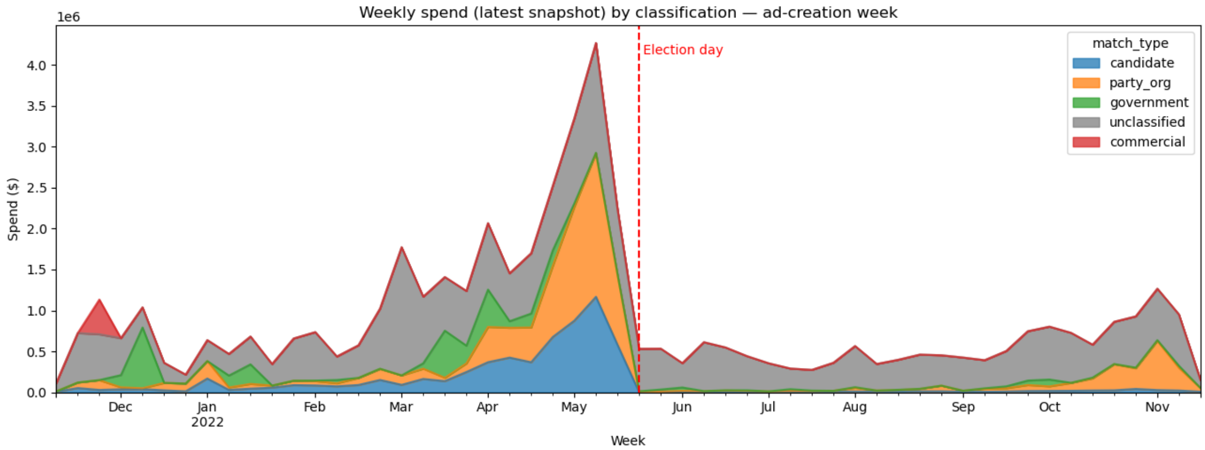


Figure 3: Weekly spend by match-type across the election window. The candidate and party advertising spike around the May 2022 election as expected; the `null` bucket shows a more diffuse pattern, which motivates the topic-clustering step in the next subsection.

2.3 Topic clustering via LDA

The non-government component contained a range of topics, not all of which were relevant to the election. The corpus was too large to label by hand and no pre-existing labels were available for supervised training. Latent Dirichlet Allocation (LDA; Blei et al., 2003) is an unsupervised topic-clustering method which can group documents by latent themes without requiring training data. It was thus well suited to this corpus. After some standard text pre-processing, LDA was fitted with 25 topics. These topics were then hand-labelled with the categories `climate`, `humanitarian_rights`, `political_advocacy`, `cost_of_living` and `mixed`. This reduced the labelling problem from hundreds of thousands of ads down to just 25 topics.

Topics without a single coherent theme were tagged as `mixed` and retained in the corpus, so downstream analysis could choose whether to include them. The result was a corpus of about 95,000 advocacy ads, written partitioned by category. The per-topic distribution is shown in Figure 4. Notably the largest spending bucket was the non-government `political_advocacy` category.

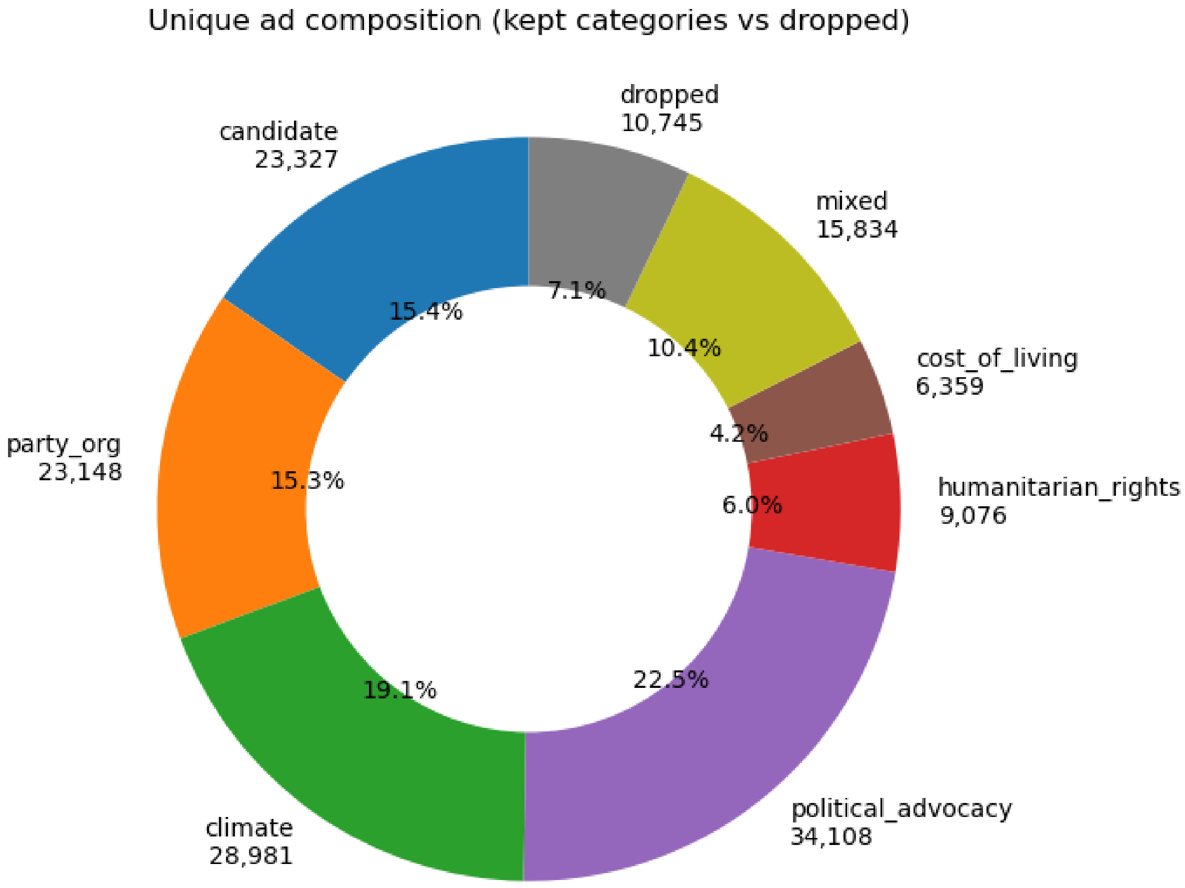


Figure 4: Distribution of ads across the 25 LDA topics with the hand-assigned categories.

2.4 Temporal analysis: persistence and centroid bands

A simple method for identifying "PAC"-style advertising is to look at spending before and after the election. An advertiser who has a high pre-election spending spike compared to the post-election period is more likely to be focused on the election.

In order to compare advertisers, spending was collated into a weekly time series per advertiser. The resulting time series was very bursty so a four-week rolling mean was then applied to smooth it. A hard boundary was also added at the election date in order to prevent bleed-over from the pre-election month into the post-election month. The resulting weekly spending pattern across the non-government corpus is shown in Figure 5.

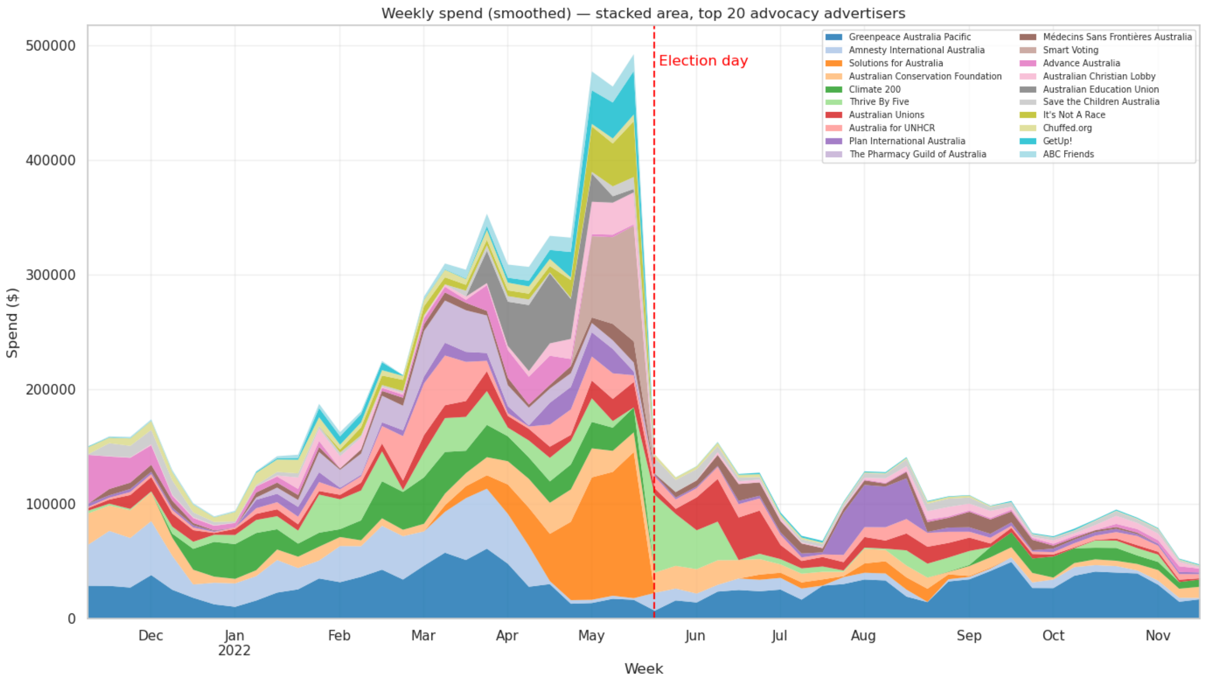


Figure 5: Weekly spend across the non-government (advocacy) corpus. Spending clusters in the weeks leading up to the May 2022 election and drops sharply afterwards.

A ratio of pre-election spend to post-election spend was calculated and saved as "persistence". Investigation of persistence found it was insufficient to identify strong pre-election spenders alone. Early-window spenders received the same weight as near-election spenders, and due to the crossover with the start of the Voice referendum, some political spenders had a late, post-election spend which reduced their ratio.

A second metric was thus developed. "Centroid" was the weighted-mean week from election day, where negative values indicate spending before the election and positive values indicate spending after. Centroid was then combined with persistence to identify advertisers who focused their spend in the near pre-election period.

Advertisers were split into four deterministic bands as shown in Table 4. The bands are also plotted on a quadrant in Figure 6.

Band	**Rule**
`election_only`	persistence < 0.30 and $\lvert\text{centroid}\rvert \leq 16$ weeks
`transitional`	$0.30 \leq$ persistence ≤ 0.80
`ongoing`	persistence > 0.80
`off_campaign`	persistence < 0.30 and $\lvert\text{centroid}\rvert > 16$ weeks

Table 4: Four-band classifier applied to top-20 advertisers.

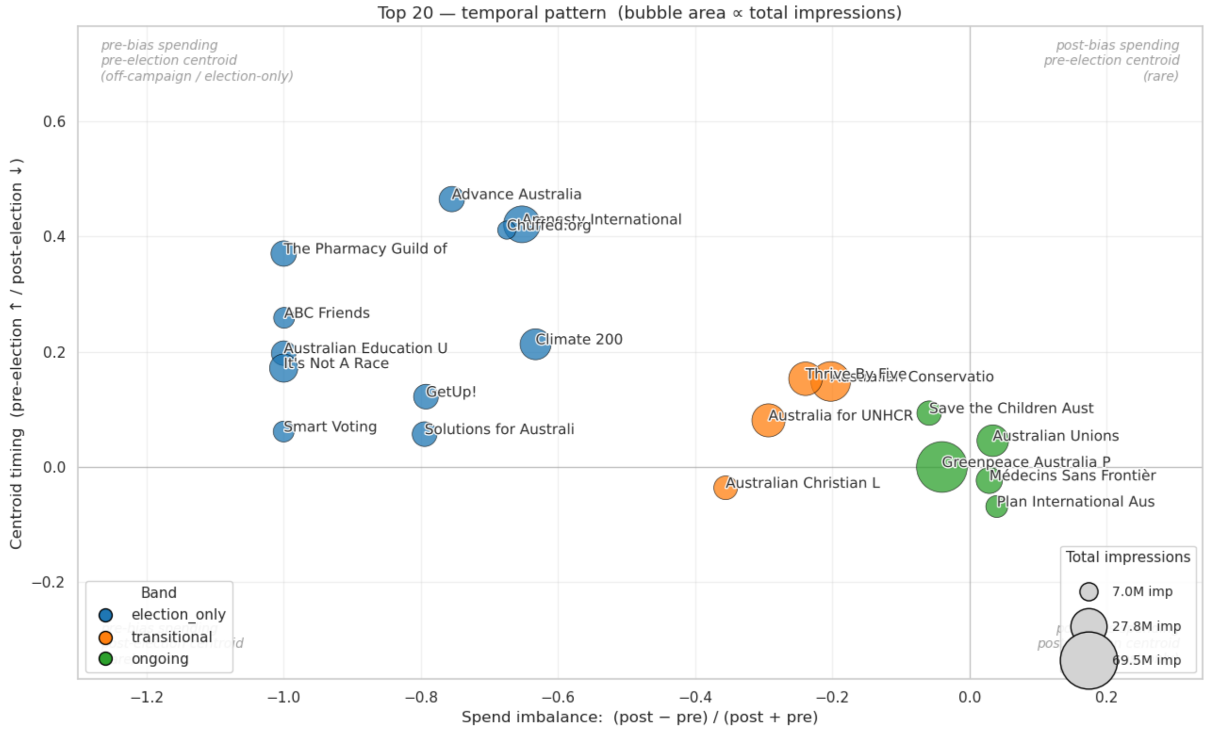


Figure 6: Top 20 non-government advertisers by spend. Low persistance and centroid indicates ongoing spend. Larger persistance and a strong negative centroid indicates an advertisiting spend focused on the pre-election period.

2.5 Content analysis: hashtag-supervised classifier

Detection based purely upon spending patterns has a number of weaknesses. Firstly, it may identify issue-based advertisers whose spend incidentally peaks before the election. Secondly, the temporal classifier reveals nothing about an advertiser's actual content. Two advertisers with identical spend patterns may be running completely different campaigns. A content-based classifier was added to capture the second case.

Supervised classification requires a training corpus, and after some experimentation hashtags proved a reliable way to extract one. About 210 political hashtags (e.g. `#federalelection`, `#morrisonmustgo`) were drawn from the `election_only` category and about 286 non-political hashtags (e.g. `#breakfreefromplastic`) from the `ongoing` category, then hand-curated. Labels were assigned as:

- Positive: at least one political hashtag, no non-political hashtag.
- Negative: at least one non-political hashtag, no political hashtag.
- Ambiguous ads excluded from training but scored at inference.

Hashtags were stripped from the text before training so the model learned from the surrounding prose rather than memorising the labels. A logistic regression classifier, a standard approach for text classification (Manning et al., 2008), was trained on the labelled corpus of around 8,000 ads.

The classifier performed strongly on held-out test data, but weaker when validated against advertisers it had not seen during training. This suggests that more advanced methods may improve coverage. Inspection of high-scoring no-hashtag ads confirmed the model caught political content the hashtag-based labelling had missed, including local-council and state-election candidates running outside the federal-election window.

Per-advertiser scores were aggregated as the mean `election_prob` across each advertiser's ads in the 4-week pre-election window. This `mean_prob_pre` metric was then exported for the combined PAC analysis in Section 3. Figure 7 shows this score across the top-20 spenders.

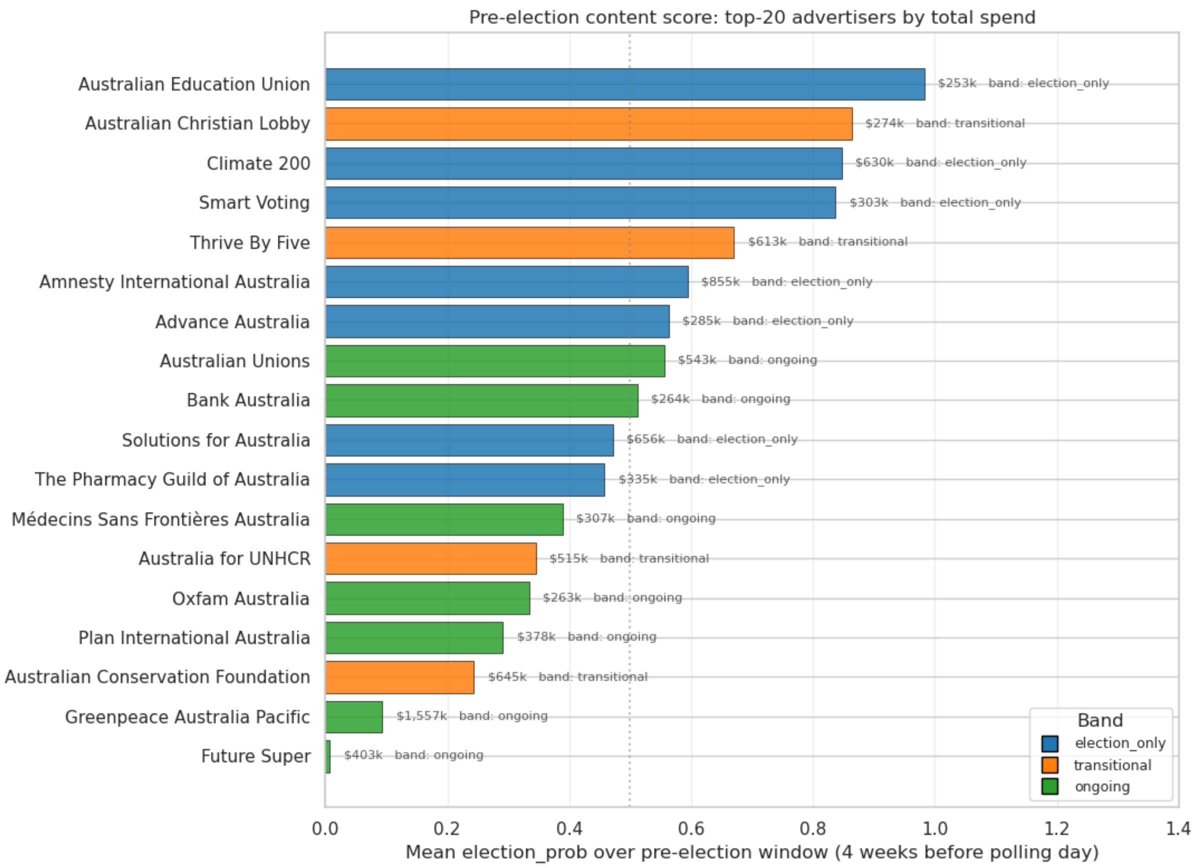


Figure 7: Mean `election_prob` over the 4-week pre-election window for the top-20 advocacy advertisers by spend, coloured by temporal band. Bars longer than 0.5 indicate advertisers whose pre-election content reads as predominantly election-themed by the classifier.

3 Discussion and Conclusions

Classifying PAC-style advertisers proved more challenging than anticipated. The temporal classifier alone tended to confuse ongoing-issue advocates such as Greenpeace and the Australian Unions with election-window spenders, while the content classifier alone picked up local-council and state-election candidates outside the federal window. Combining the two as a quadrant filter (`ec_index > 0.5` and `mean_prob_pre > 0.5`, excluding the `ongoing` band) produced a reasonable list. More advanced methods, such as transformer-based text models, may potentially improve the result.

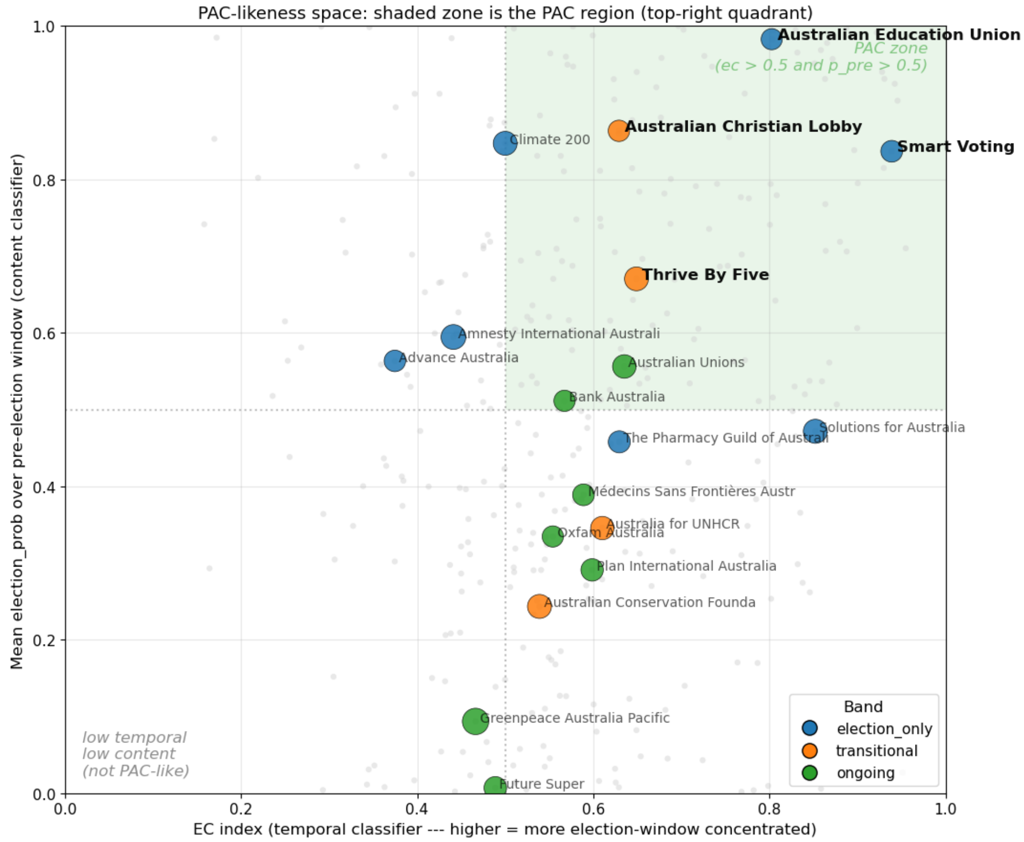


Figure 8: Combined PAC-likeness scatter: temporal classifier (`ec_index`, x-axis) against content classifier (`mean_prob_pre`, y-axis) for the top-20 advocacy advertisers. The shaded top-right quadrant is the PAC region. Greenpeace sits in the bottom-left despite the largest bubble ($1.5M).

Nevertheless several PAC-style advertisers were identified. These included Smart Voting, the Australian Education Union, Thrive By Five, the Australian Christian Lobby, GetUp!, ABC Friends, and It's Not A Race. Inspection of the top ad titles for each (Table 5) confirmed the classification. All of these advertisers combined a pre-election spend concentration with content that was explicitly partisan or directed at a specific policy outcome.

Advertiser	**Band**	**Top ad title**
Smart Voting	election_only	"Since the Coalition took power, prices have skyrocketed."
Australian Education Union	election_only	"Properly fund public schools."
Thrive By Five	transitional	"Let's make early learning and childcare fair for families."
Australian Christian Lobby	transitional	"Sign the petition to defend Christian schools."
It's Not A Race	election_only	"Government should be about setting the right priorities."
GetUp!	election_only	"Want climate leadership? Vote for it!"
ABC Friends	election_only	"Save our ABC."

Table 5: Top PAC-style advertisers identified by the combined quadrant filter, with their highest-spend ad title. All combine pre-election spend concentration with explicit issue or party-directional messaging.

The major outlier was Greenpeace. At $1.5M the largest non-government advertiser, it nevertheless satisfied none of the PAC criteria, with a fairly constant spend curve and an anti-Woodside Petroleum campaign rather than electoral content. High spend during an election window does not, on its own, indicate PAC-style behaviour.

Despite the limitations of the classifiers a reasonably confident list of Australian PAC-style advertisers can be identified from this dataset. Australia does not have formal PACs, but their informal equivalents do operate around federal elections. As discussed in the introduction, the boundary between issue advocacy and party-aligned third-party campaigning is one that traditional campaign-finance disclosure cannot easily examine. The use of the Facebook Ad Library may be one method to allow this boundary to be examined.

The scale of the dataset meant that the analysis was only practical using a distributed framework. The 40-million-row, 320 GB JSON corpus could not have been deduplicated, clustered with LDA, or joined against the candidate roll on a conventional workstation in any reasonable time. Distributed storage, lazy DataFrame evaluation, and the Spark MLlib library were all used to make the full analysis viable.

References

Armbrust, M., R. S. Xin, C. Lian, Y. Huai, D. Liu, J. K. Bradley, X. Meng, T. Kaftan, M. J. Franklin, A. Ghodsi, and M. Zaharia (2015). Spark SQL: relational data processing in Spark. In *Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*, pp. 1383–1394.

Blei, D. M., A. Y. Ng, and M. I. Jordan (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research 3*, 993–1022.

Manning, C. D., P. Raghavan, and H. Schütze (2008). *Introduction to Information Retrieval*. Cambridge, UK: Cambridge University Press.

Meng, X., J. Bradley, B. Yavuz, E. Sparks, S. Venkataraman, D. Liu, J. Freeman, D. Tsai, M. Amde, S. Owen, D. Xin, R. Xin, M. J. Franklin, R. Zadeh, M. Zaharia, and A. Talwalkar (2016). MLlib: Machine learning in Apache Spark. *Journal of Machine Learning Research 17*(34), 1–7.

Zaharia, M., R. S. Xin, P. Wendell, T. Das, M. Armbrust, A. Dave, X. Meng, J. Rosen, S. Venkataraman, M. J. Franklin, A. Ghodsi, J. Gonzalez, S. Shenker, and I. Stoica (2016). Apache spark: a unified engine for big data processing. *Communications of the ACM 59*(11), 56–65.

A Use of Generative AI

Claude (Anthropic; `claude-opus-4-7`, accessed via the Claude Code CLI between April and May 2026) was used for proofreading and editing the report text, and for help with parts of the matplotlib visualisation code in notebooks 02, 04 and 05. All substantive content, the analytical argument, the PySpark and pandas analysis pipeline, and all decisions about what to include in the report were the author's. The author reviewed and verified all code, numerical results, and claims before inclusion.

B Code

The full analysis pipeline is implemented in six Jupyter notebooks at `https://github.com/<USER>/FB_API` (replace with the actual URL). Each notebook is self-contained and writes its output as Parquet on HDFS or as CSV in the repository's `data/` directory.

`01_data_loading.ipynb` Raw JSON ingestion, schema flattening, deduplication, window filtering, write of v1 Parquet.

02_preprocessing.ipynb Candidate, party, and government byline matching via pandas UDF; write of v2 Parquet.

03_topics.ipynb LDA topic modelling, k sweep, topic inspection, output of `topic_terms.csv` for hand-labelling.

04_topic_join.ipynb Join of hand-labelled topics onto v2, filter to political-adjacent residual, write of v3 Parquet.

04b_hashtag_analysis.ipynb Regex extraction of hashtags across v3, count by band, export of `data/hashtags.csv`.

05_election_spenders.ipynb Top-20 ranking, weekly aggregation, election-boundary smoothing, persistence + centroid computation, four-band classification, export of band CSVs and visualisations.

06_election_content_classifier.ipynb Hashtag-supervised logistic-regression text classifier; FP / FN / LOO validation; per-advertiser and per-week scoring; pivot-detection table.

Selected code excerpts follow.

Election-boundary smoothing (nb 05)

```
# Hard smoothing boundary at election day --- pre and post halves
# smoothed independently so cliff-edged advertisers do not bleed forward.
weekly['is_post'] = weekly['week'] >= ELECTION_DATE
weekly['spend_smooth'] = (weekly
    .groupby(['bylines', 'is_post'])['spend']
    .transform(lambda s: s.rolling(window=4, center=True, min_periods=1).mean())
       )
```

Hashtag-supervised labelling (nb 06)

```
pol_re = r'#(' + '|'.join(re.escape(t) for t in political_hashtags) + r')\b'
non_re = r'#(' + '|'.join(re.escape(t) for t in non_political_hashtags) + r')\b'

v3 = v3.withColumn('has_political_hashtag',     col('raw_text').rlike(pol_re))
v3 = v3.withColumn('has_non_political_hashtag', col('raw_text').rlike(non_re))

# Strip hashtags before vectorising --- model must learn from surrounding prose.
v3 = v3.withColumn('text', regexp_replace(col('raw_text'), r'#\w+', ' '))

labelled = v3.withColumn('label',
    when( col('has_political_hashtag') & ~col('has_non_political_hashtag'), 1.0)
    .when(~col('has_political_hashtag') &  col('has_non_political_hashtag'),
       0.0)
    .otherwise(None))
```